Kexuan (Michael) Huang

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Education

- **University of Michigan (GPA: 3.9/4)** — Ann Arbor, MI
  M.Sc. Student in Information Science (Software Development and Big Data Analytics) — *Aug 2022 - Apr 2024 (Expected)*
  - **Graduate Student Instructor (GSI)** for Data-Oriented Programming (SI206) course
- **University of Wisconsin-Madison** — Madison, WI
  Exchange Student in Computer Science — *Jan 2022 - May 2022*
- **Shanghai Jiao Tong University (UM-SJTU Joint Institute)** — Shanghai, China
  B.Eng. in Electrical and Computer Engineering. Dual Minor in Computer Science and Entrepreneurship — *Sep 2018 - Aug 2022*
  - **Award & Scholarship**: Advanced Teaching Assistant, Merit Undergraduate Scholarship, Most Improved Scholarship

Work Experience

- **Valeo North America** — Troy, MI
  Software Engineer Intern — *Jun 2023 - Aug 2023*
  - Developed programs in C-based programming language Communication Access Programming Language (*CAPL*) for automated-parking testing. Utilized distributed systems design tool *CANoe* to validate electronic control unit (*ECU*) network signals and virtual testing environment *Vosstrex* to simulate vehicle behaviors under diverse scenarios.
  - Implement an image processing algorithm in *C++* with Open Graphics Library (*OpenGL*) to efficiently handle real-time raw video streaming from vehicle cameras placed at varying angles, ensuring adherence to strict specifications.
  - Built *Jenkins CI/CD* pipelines with *Dashing* framework for multiple projects, triggered upon codebase changes in application lifecycle management (*ALM*) system to automate project build, code analysis on *Klocwork*, testing and hardware validation.
- **Advanced Micro Devices (AMD)** — Shanghai, China
  Software Engineer Intern — *Dec 2020 - May 2021*
  - Developed a static analysis command-line tool (e.g. *VHDL*, *Verilog* and *SystemVerilog*) with *Python* for hardware description language (*HDL*) to parse source code, extract user-defined components (e.g. ports, modules and interfaces), conduct cross-validation among files and generate bug reports, which notably accelerated the routine design verification process.
  - Implemented an internal team platform with *Django* framework in *Python* on the local area network (*LAN*), which enables seamless information sharing and workflow synchronization among colleagues in office. Developed various frontend views including dashboard, to-do list, worksheet and Q&A, using *JavaScript*, *HTML*, *CSS* along with framework *Bootstrap*.
  - Revised and modernized legacy *Perl* and *Ruby* code with *Python* and *Shell* scripting, employing packages including *Requests*, *RegEx*, *Pandas*, *NumPy*, and *Sklearn* to establish a cohesive GPU test flow for efficient large-scale data process and analysis.
- **Segway-Ninebot** — Shanghai, China
  Embedded Software Engineer Intern — *Jan 2020 - Mar 2020*
  - Developed an onboard debug assistant in *C* using *FreeRTOS* for service mode of autonomous delivery robot prototype, facilitating real-time access to parameters and settings. Implemented structured folder-like view with bitmap graphic, efficient interaction logic using only few buttons, and utilized *UART*, *Bluetooth* & *Wi-Fi* for seamless log exporting solution.
  - Implemented test programs for host computers & STM micro-controllers in *C* and *Python* scripting. Performed comprehensive unit tests for communication modules and integrated tests for decision-making systems in both PC emulator and real life.

Projects

- **ChatGPT on WeChat (GitHub 550 stars, 280 forks, 1.9k users and more)**
  Open-source Project | **TypeScript, Node.js, Asynchronous Programming, Docker, CI/CD, Rapid Cloud Deployment**
  - Integrated ChatGPT into WeChat (the most popular social media in China), enabling keyword-triggered auto-reply for group & private chat, along with customized task handlers, facilitating productivity and user experience with the cutting-edge AI.
  - Leveraged *TypeScript* asynchronous programming to seamlessly handle incoming chat messages and forward responses from GPT-3.5 or GPT-4 models through the integration of *Wechaty* (an open-source project on GitHub) and the OpenAI API.
  - Released a rapid deploy template on cloud platform *Railway* by streamlining the build & deploy process with *Docker*, which currently ranks top 10 in popularity with over 1.9k users. Actively addressed and resolved over 60 issues on GitHub.
- **Intelligent Firefighting System Based on Internet-of-Things (IoT) Technology**
  Capstone Project, Full-stack Developer | **Python, TypeScript, FastAPI, React, React Native, MongoDB, Docker, CI/CD**
  - Built a fire risk management system incorporating Fuzzy Analytic Hierarchy Process (*FAHP*) and Coupling Revision to enhance accuracy and efficiency, which was successfully deployed on government servers in Shanghai for a trial run.
  - Developed the backend with *FastAPI* framework, web App with *React.js* and cross-platform mobile App for iOS & Android with *React Native*. Utilized *MongoDB NoSQL* database to handle a substantial data volume of approximately 7 million.
- **Machine-Generated Text Detection for ChatGPT**
  NLP Project, Co-author | **Python, PyTorch, Deep Learning, Large Language Model, Data Augmentation**
  - Proposed an interpretable machine-generated text detector for the ChatGPT (GPT-3.5) model, leveraging statistical analysis to attain performance on par with BERT-based models when evaluating against publicly available datasets.
  - Present data augmentation techniques for various downstream tasks like Q&A systems and text completion, by analyzing the acquired discriminator and presenting the weights & attention visualizations.

Skills

- **Programming**: Verilog, Assembly (x86), C, C++, Java, Python, MATLAB, TypeScript, HTML, SQL, R, Shell, LaTeX
- **Development**: Git, GitHub, Docker, RESTful, React, Django, Hadoop, Spark, Drill, Vim, VS Code, macOS, Linux
- **Specialty**: Badminton (former professional athlete, current UMich team player), Anime Piano